

L10a: Monte Carlo Pricing and Sensitivity Analysis of American Options Premium

CHEME 5660 Cornell University

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Objectives for Today

Previously, we priced American options on a CRR tree. Can simulation price them? Duration and convexity measured how a bond's price moves with its yield. How much does the premium change when the share price, volatility, time to expiration, or interest rate changes?

Today's concepts

- Estimate and decompose the premium of American options: Longstaff–Schwartz Monte Carlo, then intrinsic value (immediate exercise value) and extrinsic value (time value).
- Compute the Greeks for American options: Delta, Theta, Vega, Rho, and Gamma from a local Taylor expansion of the options premium.
- Interpret Delta and compute the probabilities of exercise and profit: the shares needed to hedge an option position, and the probabilities from projected share price distributions at the strike price and at the breakeven.

Lectures and Examples

Lecture:
CHEME-5660-L10a-Lecture-American-Derivatives-Greeks-Fall-2026.ipynb

Example 1:
CHEME-5660-L10a-Example-LSM-Premium-Fall-2026.ipynb

Example 2:
CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

Example 3:
CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Review on your own:
CHEME-5660-L10a-Example-ExtrinsicValue-Fall-2026.ipynb

Company Profile: Susquehanna

Susquehanna (SIG) is a privately held trading firm whose core business is options trading. Friends who traded options on the floor of the Philadelphia Stock Exchange started it in 1987.

- Options market making: it posts prices to buy and sell, so it takes the other side of its customers' contracts. In October 2026, it was the Designated Primary Market Maker for 1,359 of 5,370 Cboe option classes, the most of any firm.
- Managing the Greeks: its book is exposed to Delta, Vega, Theta, and Rho at once, and the goal is to net these across products.
- Training: interns learn options theory, and poker teaches new traders to decide under uncertainty.

Connection to today's lecture: when a customer buys a call, the firm hedges with $100\,\Delta_c$ shares and rebalances as Gamma moves Δ_c.

Monte Carlo Pricing of American Options

We simulate share price paths under the risk-neutral pricing rule $\mathbb{Q}$, then estimate the value of holding at each date by regressing the discounted cash flows from holding on the current share price across paths. Like the lattice, Longstaff and Schwartz work backward from expiration:

Initialize: given the share price S_0, the strike price K, the payoff function $V(K, S)$ of the call or put, the time to expiration T (years), the volatility σ, the risk-free rate g_y, the number of exercise dates h after today, and the number of paths M. Compute $\Delta t = T/h$ and $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

Reference: Longstaff, F. A., and Schwartz, E. S. (2001). Valuing American options by simulation: a simple least-squares approach. Review of Financial Studies 14(1): 113–147.

Pseudocode: Longstaff–Schwartz

1. Simulate: generate M share price paths under $\mathbb{Q}$ from S_0 on the dates $t = 0, 1, \ldots, h$ (expiration), where $S_i(t)$ is the share price on path i at date t.
2. At expiration, set each path's cash flow to its payoff: $Y_i(h) \leftarrow V(K, S_i(h))$ for $i = 1, 2, \ldots, M$, where $Y_i(t)$ is path i's cash flow valued at date t.
3. For each date $t = h-1, h-2, \ldots, 1$ **do**:
   a. Discount each cash flow back one date: $Y_i(t) \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot Y_i(t+1)$.
   b. Regress: on the in-the-money paths, those with $V(K, S_i(t)) > 0$, fit the value of holding $\hat{C}_t(S) = \beta_0 + \beta_1 S + \beta_2 S^2$ to the pairs $(S_i(t), Y_i(t))$ by least squares.
   c. Decide: for each in-the-money path, if $V(K, S_i(t)) > \hat{C}_t(S_i(t))$, **then**: exercise, so $Y_i(t) \leftarrow V(K, S_i(t))$.
4. Discount the cash flows to today and average them: $\bar{Y} \leftarrow \mathcal{D}_{1,0}^{-1}(g_y) \cdot \frac{1}{M} \sum_{i=1}^{M} Y_i(1)$.
5. The premium is $\mathcal{P}^{\text{AM}} \leftarrow \max\{V(K, S_0),\ \bar{Y}\}$.

The Early-Exercise Premium by Simulation

Let's price an American put by simulation and compare it with the lattice and the European premium. We apply two rules to the same simulated paths:

Rule at each date	Estimates
Longstaff–Schwartz: exercise when $V(K, S_i(t)) > \widehat{C}_t(S_i(t))$	$\mathcal{P}^{\text{AM}}$
Hold on every path	$\mathcal{P}^{\text{EU}}$

$$\boxed{\text{early-exercise premium} \approx \mathcal{P}^{\text{AM}} - \mathcal{P}^{\text{EU}}}$$

Example 1: CHEME-5660-L10a-Example-LSM-Premium-Fall-2026.ipynb

Extrinsic versus Intrinsic Value

The premium (price) of an American option can be broken down into two components: intrinsic value and extrinsic value.

- Intrinsic value: the immediate exercise value, $\max(0, S(t) - K)$ for a call and $\max(0, K - S(t))$ for a put, where $S(t)$ is the current share price.
- Extrinsic value: also known as time value, the additional value from the potential for future price movements before expiration.

The extrinsic value $E_V(t)$ can be expressed mathematically as:

$$E_V(t) = \text{Option Price} - \text{Intrinsic Value} \quad \blacksquare$$

Extrinsic Value

In expanded form, the extrinsic value $E_V(t)$ is:

$$E_V(t) = \begin{cases} \mathcal{P}_c(K, S(t)) - \max(0, S(t) - K) & \text{(for Call Options)} \\ \mathcal{P}_p(K, S(t)) - \max(0, K - S(t)) & \text{(for Put Options)} \end{cases}$$

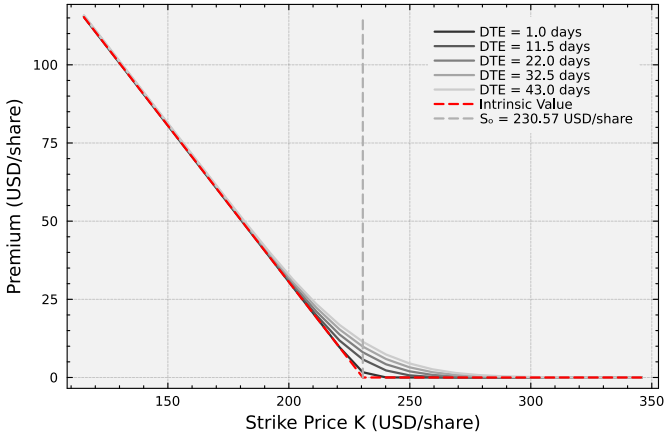
When the option is out of the money, it has no intrinsic value, and the entire option price is extrinsic value.

Why do people pay money for something that is "worthless"? There is still a chance that the share price moves favorably before expiration.

As time moves towards expiration, the extrinsic value tends to decrease (time decay), because there is less time for the share price to move favorably.

Review on your own: CHEME-5660-L10a-Example-ExtrinsicValue-Fall-2026.ipynb

Call Premium across Strikes and Expirations



Hmmmm: the premium changes with time to expiration and the strike price. How does it change with other factors?

The Greeks

The Greeks are local sensitivity coefficients that measure how American options premiums change with market conditions:

- Delta Δ: the change in premium for a `+1 USD/share` change in the share price.
- Theta Θ: the change in premium for a `1-day decrease` in the lifetime of the contract (time decay).
- Vega ν: the change in premium for a `+1%` change in volatility.
- Gamma Γ: the change in Δ for a `+1 USD/share` change in the share price.
- Rho ρ: the change in premium for a `+1%` change in the risk-free rate, usually minor except for longer-term options.

Most trading platforms compute these for you!

A Local Taylor Expansion of the Premium

Expanding the premium $\mathcal{P}_\star(S, \sigma, t, g_y)$ of a contract of type $\star \in \{c, p\}$ at calendar time t to second order in S and first order in the other inputs:

$$\boxed{\Delta\mathcal{P}_\star \approx \Delta_\star\,\Delta S + \nu_\star\,\Delta\sigma + \Theta_\star\,\Delta t + \rho_\star\,\Delta g_y + \tfrac{1}{2}\,\Gamma_\star\,(\Delta S)^2}$$

$$\Delta_\star = \frac{\partial \mathcal{P}_\star}{\partial S} \qquad \nu_\star = \frac{\partial \mathcal{P}_\star}{\partial \sigma} \qquad \Theta_\star = \frac{\partial \mathcal{P}_\star}{\partial t} \qquad \rho_\star = \frac{\partial \mathcal{P}_\star}{\partial g_y} \qquad \Gamma_\star = \frac{\partial^2 \mathcal{P}_\star}{\partial S^2}$$

Derivatives are taken at the current inputs, with Δt in years and $\Delta\sigma$, Δg_y in decimals. Platforms quote $\Theta_\star/365$ per day, and $\nu_\star/100$ and $\rho_\star/100$ per percentage point.

Delta and Gamma as Duration and Convexity

$\Delta_\star$ and $\Gamma_\star$ play the roles of a bond's duration and convexity: the slope and the curvature of the premium as a function of the share price.

Example: a long call has $\Delta_c = 0.50$ and $\Gamma_c = 0.02$ per USD/share.

$$\Delta S = +5: \quad \Delta \mathcal{P}_c \approx 0.50 \times 5 + \tfrac{1}{2}(0.02)(5)^2 = 2.50 + 0.25 = 2.75 \text{ USD/share}$$

$$\Delta S = -5: \quad \Delta \mathcal{P}_c \approx -2.50 + 0.25 = -2.25 \text{ USD/share}$$

The Gamma term adds to the premium whichever way the share price moves.

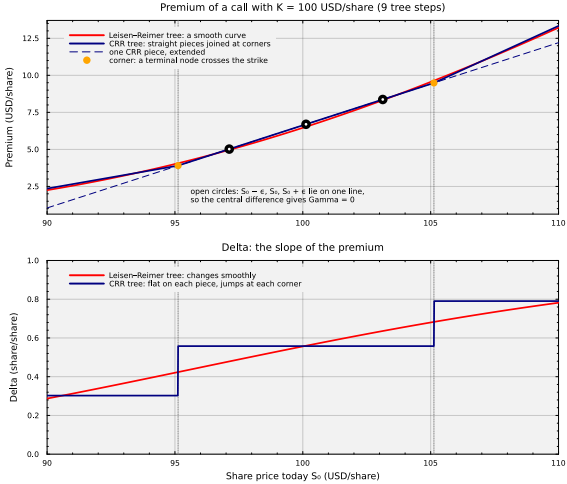
Computing the Greeks of American Options

Backward induction on the CRR tree gives no formula to differentiate, so we reprice the tree with one input moved up and down by a step ϵ:

$$\frac{\partial \mathcal{P}_\star}{\partial x} \approx \frac{\mathcal{P}_\star(x+\epsilon) - \mathcal{P}_\star(x-\epsilon)}{2\epsilon} \qquad x \in \{S, \sigma, t, g_y\}$$

$$\Gamma_\star \approx \frac{\mathcal{P}_\star(S+\epsilon) - 2\,\mathcal{P}_\star(S) + \mathcal{P}_\star(S-\epsilon)}{\epsilon^2}$$

What Goes Wrong on the CRR Tree?



Corners: the CRR factors do not depend on S_0, so the premium is straight pieces joined where a terminal node crosses the strike (top).

One piece: a step ϵ between two corners sees one piece (open circles). Delta is its slope, and Gamma is zero.

Jumps: Delta jumps at each corner (bottom). More tree steps move the corners closer but never remove them.

Leisen and Reimer remove the corners with a tree built around the strike.

Reference: Leisen and Reimer (1996). Applied Mathematical Finance 3(4): 319–346.

The Leisen–Reimer Tree

Leisen and Reimer choose q so the tree finishes above the strike as often as GBM does under $\mathbb{Q}$:

$$
\begin{aligned}
S_T &= S_0 \exp\left[\left(g_y - \tfrac{\sigma^2}{2}\right) T + \sigma\sqrt{T}\, Z\right] && \text{GBM under } \mathbb{Q}, \text{ with } Z \sim \mathcal{N}(0,1) \\
\ln S_T &= \ln S_0 + \left(g_y - \tfrac{\sigma^2}{2}\right) T + \sigma\sqrt{T}\, Z && \text{take the logarithm} \\
\mathbb{Q}\left(S_T > K\right) &= \mathbb{Q}\left(\ln S_T > \ln K\right) && \ln \text{ is increasing} \\
&= \mathbb{Q}\left(\ln S_0 + \left(g_y - \tfrac{\sigma^2}{2}\right) T + \sigma\sqrt{T}\, Z > \ln K\right) && \text{substitute } \ln S_T \\
&= \mathbb{Q}\left(Z > \frac{\ln K - \ln S_0 - \left(g_y - \sigma^2/2\right) T}{\sigma\sqrt{T}}\right) && \text{subtract, then divide by } \sigma\sqrt{T} > 0 \\
&= \mathbb{Q}\left(Z > -d_-\right) && \ln K - \ln S_0 = -\ln(S_0/K) \\
&= 1 - N(-d_-) && N \text{ is the standard normal CDF} \\
\implies & \boxed{\mathbb{Q}\left(S_T > K\right) = N(d_-)} && N(-z) = 1 - N(z)
\end{aligned}
$$

where $d_- = \left[\ln(S_0/K) + \left(g_y - \sigma^2/2\right) T\right] / (\sigma\sqrt{T})$. This probability depends only on the share price, so it holds for American contracts too.

Leisen–Reimer Probabilities and Factors

With an odd number of steps h, the tree is centered on the strike and finishes above it when more than half of its moves are up. The Peizer–Pratt inversion $H(z;h)$ gives the up probability that makes this happen with probability close to $N(z)$:

$$H(z;h) = \frac{1}{2} + \frac{\operatorname{sgn}(z)}{2}\sqrt{1 - \exp\left[-\left(\frac{z}{h + 1/3 + 0.1/(h+1)}\right)^2 \left(h + \frac{1}{6}\right)\right]}$$

$$q = H(d_-;h) \qquad q' = H(d_+;h) \qquad \text{risk-neutral and second probabilities}$$

$$u = e^{g_y \Delta t}\,\frac{q'}{q} \qquad d = e^{g_y \Delta t}\,\frac{1-q'}{1-q} \qquad \text{up and down factors}$$

where $d_+ = d_- + \sigma\sqrt{T}$ and $\Delta t = T/h$. Like the CRR factors, they grow the expected share price at the risk-free rate:

$$qu + (1-q)d = q\,e^{g_y \Delta t}\frac{q'}{q} + (1-q)\,e^{g_y \Delta t}\frac{1-q'}{1-q} = e^{g_y \Delta t}\,(q' + 1 - q') = e^{g_y \Delta t}$$

No corners: $d_\pm$ depend on S_0, and so do the factors. The strike stays between the two middle terminal nodes as S_0 moves.

Greeks on the Leisen–Reimer Tree

Initialize: given S_0, K, the payoff function $V(K, S)$ of the call or put, T (years), σ, g_y, an odd number of time steps h, and a step ϵ_x for each input $x \in \{S_0, \sigma, T, g_y\}$, in the units of that input. Let $\mathcal{P}(S_0, \sigma, T, g_y)$ be the premium that steps 1–4 compute from a set of inputs.

1. Compute $\Delta t \leftarrow T/h$, then $d_- \leftarrow \left[\ln(S_0/K) + \left(g_y - \sigma^2/2\right) T\right] / (\sigma\sqrt{T})$ and $d_+ \leftarrow d_- + \sigma\sqrt{T}$.
2. Compute the probabilities $q \leftarrow H(d_-; h)$ and $q' \leftarrow H(d_+; h)$.
3. Compute the factors $u \leftarrow e^{g_y \Delta t}\, q'/q$ and $d \leftarrow e^{g_y \Delta t}\, (1 - q')/(1 - q)$.
4. Compute the premium with the CRR pricing pseudocode, using these u, d, and q: $\mathcal{P}(S_0, \sigma, T, g_y) \leftarrow \hat{V}_{0,0}$.

Pseudocode: Leisen–Reimer Greeks

5. For each input $x \in \{S_0, \sigma, T, g_y\}$ **do**:
   a. Compute $\mathcal{P}(x + \epsilon_x)$ and $\mathcal{P}(x - \epsilon_x)$ with steps 1–4, moving only x. The factors depend on every input, so each evaluation rebuilds the tree.
   b. Compute the central difference: $\partial\mathcal{P}/\partial x \leftarrow [\mathcal{P}(x + \epsilon_x) - \mathcal{P}(x - \epsilon_x)]\,/(2\epsilon_x)$.
6. Compute $\Delta \leftarrow \partial\mathcal{P}/\partial S_0$, $\nu \leftarrow \partial\mathcal{P}/\partial\sigma$, $\rho \leftarrow \partial\mathcal{P}/\partial g_y$, and $\Theta \leftarrow -\partial\mathcal{P}/\partial T$, because calendar time moves forward as the time to expiration falls.
7. Compute Gamma from the three share price evaluations: $\Gamma \leftarrow \left[\mathcal{P}(S_0 + \epsilon_{S_0}) - 2\,\mathcal{P}(S_0) + \mathcal{P}(S_0 - \epsilon_{S_0})\right]/\epsilon_{S_0}^2$.

Output: the premium $\mathcal{P}(S_0, \sigma, T, g_y)$ and the Greeks Δ, Γ, Θ, ν, and ρ.

Before reporting a Greek, recompute it with about twice as many time steps, still odd, and smaller steps ϵ_x. Report it once it stops changing.

Example 2: CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

The Secret Lives of Delta

Besides measuring the premium change for a `+1 USD/share` move in the share price, Δ has two other interpretations:

- Hedge ratio: for every n contracts held ($n < 0$ if the contracts are sold), hold $-100 \cdot n \cdot \Delta$ shares to be delta-neutral, i.e., insensitive to small share price moves.
- Probability of exercise: Δ is often used as an estimate of the probability that the option expires in-the-money. A call with $\Delta = 0.7$ suggests a 70% chance.

Probabilities of Exercise and Profit

Probability of exercise (POE): the probability that the contract finishes in-the-money at expiration $t = T$:

$$\text{POE} = \begin{cases} \mathbb{P}\left(S_T > K + 0.01\right) & \text{(for Call Options)} \\ \mathbb{P}\left(S_T < K - 0.01\right) & \text{(for Put Options)} \end{cases}$$

where S_T is the share price at expiration and K is the strike price.
Conjecture: $\Delta \approx$ POE. Example 3 checks it.

Probability of profit (POP): the probability that the contract's profit at expiration is positive:

$$\text{POP} = \mathbb{P}\left(\hat{P}_{\mathcal{C}} > 0\right)$$

where $\hat{P}_{\mathcal{C}}$ is the profit of the (single or composite) contract at expiration.

Computing POE and POP by Simulation

POE and POP are forecasts, so we compute them under our belief $\mathbb{P}$: GBM with our estimates of the drift μ and the volatility σ.

Initialize: given S_0, K, the premium $\mathcal{P}$ of the call or put, T (years), μ and σ under $\mathbb{P}$, g_y, and the number of samples M. Compute the breakeven share price $S^\ddagger = K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}$ for a call, or $S^\ddagger = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}$ for a put, where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$.

1. For each sample $i = 1, 2, \ldots, M$ **do**:
   a. Draw a sample from the standard normal distribution: $Z_i \sim \mathcal{N}(0,1)$.
   b. Compute the share price at expiration: $S_i(T) \leftarrow S_0 \exp\left[(\mu - \sigma^2/2)\,T + \sigma\sqrt{T}\,Z_i\right]$.
2. Compute the cumulative distribution function, the fraction of samples at or below each share price x: $\widehat{F}(x) \leftarrow \frac{1}{M} \cdot \#\{i : S_i(T) \le x\}$.

Pseudocode: POE and POP

3. Compute the probability of exercise. If the contract is a call, **then** POE $\leftarrow 1 - \hat{F}(K + 0.01)$. Otherwise, POE $\leftarrow \hat{F}(K - 0.01)$.
4. Compute the probability of profit:
   - If we bought a call or sold a put, **then** the profit is positive above the breakeven: POP $\leftarrow 1 - \hat{F}(S^{\ddagger})$.
   - If we sold a call or bought a put, **then** the profit is positive below the breakeven: POP $\leftarrow \hat{F}(S^{\ddagger})$.
   - If the contract is composite and its profit changes sign at more than one share price, **then** count the profitable samples: POP $\leftarrow \frac{1}{M} \cdot \#\left\{i : \hat{P}_{\mathcal{C}}\left(S_i(T)\right) > 0\right\}$.

Output: the probability of exercise POE and the probability of profit POP.

With $\mu = g_y$ and σ set to the implied volatility, the same steps give the probabilities under $\mathbb{Q}$. These are the probabilities inside the market price, not our forecast.

Why Is Delta Close to the POE?

With no dividends, a call's American and European premiums are equal. Unrolling the European backward induction on the CRR tree:

$$\mathcal{P}_c = \mathcal{D}_{T,0}^{-1}(g_y) \sum_{k=0}^{h} \binom{h}{k} q^k (1-q)^{h-k} \max\left(S_0 u^k d^{h-k} - K,\ 0\right)$$

The factors do not depend on S_0, so a payoff has slope $u^k d^{h-k}$ in S_0 if its node is in the money, and zero otherwise. Define $q^* = q\,u\,e^{-g_y \Delta t}$. Dividing $qu + (1-q)d = e^{g_y \Delta t}$ by $e^{g_y \Delta t}$ gives $(1-q)\,d\,e^{-g_y \Delta t} = 1 - q^*$, so q^* is also a probability. Then Delta is:

$$\Delta_c = \frac{\partial \mathcal{P}_c}{\partial S_0} = \mathcal{D}_{T,0}^{-1}(g_y) \sum_{k \in \text{ITM}} \binom{h}{k} q^k (1-q)^{h-k}\, u^k d^{h-k} \qquad \text{only ITM payoffs have a slope}$$

$$= \sum_{k \in \text{ITM}} \binom{h}{k} \left(q\,u\,e^{-g_y \Delta t}\right)^k \left((1-q)\,d\,e^{-g_y \Delta t}\right)^{h-k} \qquad \mathcal{D}_{T,0}^{-1}(g_y) = e^{-g_y h \Delta t}\text{, split over } h \text{ steps}$$

$$= \sum_{k \in \text{ITM}} \binom{h}{k} \left(q^*\right)^k \left(1-q^*\right)^{h-k} \qquad \text{substitute } q^* \text{ and } 1-q^*$$

where ITM is the set of in-the-money terminal nodes.

Delta Is a Probability of Exercise

The risk-neutral POE of the call is the same sum with q in place of q^*:

$$\Delta_c = \sum_{k \in \text{ITM}} \binom{h}{k} (q^*)^k (1-q^*)^{h-k} \qquad \mathbb{Q}(S_T > K) = \sum_{k \in \text{ITM}} \binom{h}{k} q^k (1-q)^{h-k}$$

- Delta is larger than the risk-neutral POE: $u > e^{g_y \Delta t}$, so $q^* > q$. On the Leisen–Reimer tree, q^* is the second probability q'.
- When is Delta close? Each step adds about $\frac{1}{2}\sigma\sqrt{\Delta t}$ to the up probability, so the gap grows with $\sigma\sqrt{T}$. Delta is close for short-dated contracts.
- Our forecast and American puts: under $\mathbb{P}$, the up probability is the real-world p, so the forecast POE can sit above or below Delta. Early exercise adds to $|\Delta_p|$, so the match is approximate.

Example 3: CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Summary

Today we priced American options by simulation, split the premium into intrinsic and extrinsic value, and introduced the Greeks.

Key takeaways

- Estimating and decomposing the American premium: Longstaff–Schwartz regresses the value of holding on the share price. Intrinsic value is the immediate exercise value, and extrinsic value is the potential for favorable price movement.
- The Greeks quantify option price sensitivity: they are the partial derivatives in a local Taylor expansion of the premium, which we estimate for American options by repricing a binomial tree. The Leisen–Reimer tree avoids the corners of the CRR premium.
- Delta serves three purposes: the rate of change in the premium, the hedge ratio for delta-neutral positions, and an approximation of the POE. On the tree, a call's Delta is exactly a POE computed with a slightly larger up probability. POP reads the projected share price distribution at the breakeven.